A.2 Figure 2 details

Figure 2 panels (B-D) show the allowed regions of Procrustes distance and expected Euclidean distance between decoded signals for representations normalized to have $\operatorname{Tr} \tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^\top = \operatorname{Tr} \tilde{\boldsymbol{Y}}\tilde{\boldsymbol{Y}}^\top = 1$, with varying participation ratio $\mathcal{R}_\Delta$. We have populated the plots with points representing the respective distances between randomly sampled pairs of positive semi-definite matrices. These positive semi-definite matrices represent normalized kernel matrices $\tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^\top$ and $\tilde{\boldsymbol{Y}}\tilde{\boldsymbol{Y}}^\top$. We have seen in the main text that the squared Euclidean distance between these kernel matrices is equivalent to the expected squared Euclidean distance between decoded signals, $\mathbb{E}\|\boldsymbol{X}\boldsymbol{w}^* - \boldsymbol{Y}\boldsymbol{v}^*\|_2^2 = \|\tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^\top - \tilde{\boldsymbol{Y}}\tilde{\boldsymbol{Y}}^\top\|_F^2$, when the distribution of decoding targets satisfies $\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top] = \boldsymbol{I}$. On the other hand, [16] demonstrated how the Procrustes distance $\mathcal{P}(\tilde{\boldsymbol{X}}, \tilde{\boldsymbol{Y}})$ is equivalent to the Bures distance eq. (27), $d_B(\tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^\top, \tilde{\boldsymbol{Y}}\tilde{\boldsymbol{Y}}^\top)$. Therefore, both distances on these plots can be calculated from samples of positive semi-definite matrix pairs.

Each color of points in fig. 2 (B-D) represents a different random ensemble of $M \times M$ PSD matrices with $M = 50$, which are then binned into one of the three plots if the participation ratio $\mathcal{R}_\Delta$ lies in the respective range indicated above each plot panel.

- Pink points are generated by sampling eigenvalues for the two PSD matrices from a Dirichlet distribution with concentration parameters logarithmically spaced between 10^{-3} and 10^3. Matrices of eigenvectors are then randomly sampled from the set of orthogonal matrices.
- Green points were generated by sampling a random matrix $\boldsymbol{A}$ of size $M \times r$ with each element drawn from the uniform distribution $\mathcal{U}[0, 1]$. A uniform random integer r was selected between 1 and 24. A PSD matrix was generated by multiplying $\boldsymbol{A}\boldsymbol{A}^\top$. Another PSD matrix was then generated by adding a randomly weighted matrix of the same dimension with standard normal distributed entries $\boldsymbol{N}$ to $\boldsymbol{A}$, so $\boldsymbol{B} = \boldsymbol{A} + \epsilon\boldsymbol{N}$, where $\epsilon \sim \mathcal{U}[0, 1]$. The distance between $\frac{1}{\operatorname{Tr}\boldsymbol{A}\boldsymbol{A}^\top}\boldsymbol{A}\boldsymbol{A}^\top$ and $\frac{1}{\operatorname{Tr}\boldsymbol{B}\boldsymbol{B}^\top}\boldsymbol{B}\boldsymbol{B}^\top$ is then computed using these two distance metrics.
- Blue points are seen to only occupy fig. 2 (B), as these correspond to distances between matrices of rank 1 and trace 1. As we saw earlier in this appendix, in this case the participation ratio of the difference between these two PSD matrices is always 2. The rank 1 matrices were sampled using the same method as the green points above, but setting $r = 1$.
- Purple points were generated by sampling one matrix from a Wishart distribution $W_M(r, \boldsymbol{I})$, with degrees of freedom r chosen as a uniformly random integer between 1 and 50. This matrix was normalized to have trace 1. The second matrix was generated by adding to the first matrix another matrix drawn from the Wishart distribution $W_M(n, \boldsymbol{I})$, with n a randomly selected integer between 1 and 10. Both matrices are normalized to have trace 1.
- Yellow points represent distances between PSD matrices that are simultaneously diagonalizable. These were generated by sampling M values uniformly between 0 and 1 for each matrix, but and setting these values to 0 if they fall below the threshold 0.6. The resultant values were then normalized to sum to 1, which became the eigenvalues for the two PSD matrices. A matrix of eigenvectors was then randomly selected from the set of orthogonal matrices.

A.3 Generalizations and Interpretations as $M \to \infty$

In the main text, we considered quantifying representational similarity across a finite set of M stimulus conditions. Further, when considering the average distance in decoding performance, we have assumed that $\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top] = \boldsymbol{I}$. Here, we briefly discuss how to relax both constraints, leaving a full investigation to future work. Readers will need familiarity with the basic principles of kernel methods and gaussian processes in machine learning (see e.g. [31, 36]) to follow along with certain results in this section.

To begin, we must revisit and refine our theoretical framework developed in section 2. Under proposition 2, we treated the neural response matrices, $\boldsymbol{X}$ and $\boldsymbol{Y}$, as constants while we treated the decoder targets, $\boldsymbol{z}$, as a random variable. In this section we will treat $\boldsymbol{X}$, $\boldsymbol{Y}$ and $\boldsymbol{z}$ as joint random variables, as has been done in prior work [27, 3]. Specifically, let $\mathcal{U}$ denote the space

of possible stimulus inputs (e.g. the space of natural images or grammatically correct English sentences). We are interested in quantifying similarity between two neural networks, $f : \mathcal{U} \mapsto \mathbb{R}^{N_X}$ and $g : \mathcal{U} \mapsto \mathbb{R}^{N_Y}$, across this stimulus space. Let $P_{\boldsymbol{u}}$ denote a distribution with support on $\mathcal{U}$, and let $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_M$ denote independent samples from $P_{\boldsymbol{u}}$. As before, we collect the network responses into matrices $\boldsymbol{X} \in \mathbb{R}^{M \times N_X}$ and $\boldsymbol{Y} \in \mathbb{R}^{M \times N_Y}$, which we assume to be mean centered, $\mathbb{E}[f(\boldsymbol{u})] = \mathbb{E}[g(\boldsymbol{u})] = \boldsymbol{0}$. The rows of these matrices are given by $\boldsymbol{x}_i = f(\boldsymbol{u}_i)$ and $\boldsymbol{y}_i = g(\boldsymbol{u}_i)$, and we assume that $\|f(\boldsymbol{u}_i)\|_2 \leq \infty$ for all inputs indexed by $i \in \{1, ..., M\}$. This is a common assumption that can be interpreted in the neuroscience sense as assuming neural firing rates are bounded from above (they cannot become arbitrarily large).

Finally, we define a decoding task as a function $\eta : \mathcal{U} \mapsto \mathbb{R}$. For M finite samples (as considered in the main text), the decoder target is the vector with elements consisting of η evaluated at each sampled input: $\boldsymbol{z} = [\eta(\boldsymbol{u}_1) \quad \ldots \quad \eta(\boldsymbol{u}_M)]^\top$. As in sections 3 and 4, we will be interested in characterizing the *average* decoder alignment over multiple tasks. In the main text we considered $\mathbb{E}_{\boldsymbol{z}} \langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle$, but we would like to now generalize this to an expectation of an inner product between two *functions* over the input space. We consider η to be drawn from a Gaussian process, $\eta \sim \mathcal{GP}(q)$, with a covariance operator $q : \mathcal{U} \times \mathcal{U} \mapsto \mathbb{R}$. Intuitively, q defines the difficulty of the distribution over regression tasks by specifying smoothness with respect to the input space $\mathcal{U}$. For example, the popular squared exponential or radial basis function (RBF) kernel:

$$q(\boldsymbol{u}, \boldsymbol{u}') = \exp\left(-\frac{\|\boldsymbol{u} - \boldsymbol{u}'\|_2^2}{2\gamma}\right) \tag{54}$$

comes equipped with a length scale parameter $\gamma > 0$ that determines the smoothness of functions sampled $\eta \sim \mathcal{GP}(q)$. In the limit that $\gamma \to 0$ we get the Kronecker delta function:

$$\delta(\boldsymbol{u}, \boldsymbol{u}') = \begin{cases} 1 & \boldsymbol{u} = \boldsymbol{u}' \\ 0 & \boldsymbol{u} \neq \boldsymbol{u}' \end{cases} \tag{55}$$

In the remainder of this section, we show that the average decoding similarity converges to an expected inner product between kernels defined by the two neural systems. Specifically, let us define $k : \mathcal{U} \times \mathcal{U} \mapsto \mathbb{R}$ and $h : \mathcal{U} \times \mathcal{U} \mapsto \mathbb{R}$ as follows:

$$k(\boldsymbol{u}, \boldsymbol{u}') = f(\boldsymbol{u})^\top \boldsymbol{G}(f)^{-1} f(\boldsymbol{u}') \qquad \text{and} \qquad h(\boldsymbol{u}, \boldsymbol{u}') = g(\boldsymbol{u})^\top \boldsymbol{G}(g)^{-1} g(\boldsymbol{u}') \tag{56}$$

where we have generalized eq. (4) as:

$$\boldsymbol{G}(f) = a\mathbb{E}[f(\boldsymbol{u}) f(\boldsymbol{u})^\top] + b\boldsymbol{I} \tag{57}$$

We denote by $L^2(\mathcal{U})$ the space of L^2-integrable functions from the set $\mathcal{U}$ to $\mathbb{R}$. Consider the integral operators $\mathcal{K} : L^2(\mathcal{U}) \to L^2(\mathcal{U})$ and $\mathcal{H} : L^2(\mathcal{U}) \to L^2(\mathcal{U})$ associated with the kernels in eq. (56). We define the functions $\eta_{\mathcal{K}}(\boldsymbol{u})$ and $\eta_{\mathcal{H}}(\boldsymbol{u})$ as these operators acting on the decoding task function $\eta(\boldsymbol{u})$:

$$[\mathcal{K}\eta](\boldsymbol{u}) = \int k(\boldsymbol{u}, \boldsymbol{u}') \eta(\boldsymbol{u}') p(\boldsymbol{u}') d\boldsymbol{u}' \equiv \eta_{\mathcal{K}}(\boldsymbol{u}) \tag{58}$$

$$[\mathcal{H}\eta](\boldsymbol{u}) = \int h(\boldsymbol{u}, \boldsymbol{u}') \eta(\boldsymbol{u}') p(\boldsymbol{u}') d\boldsymbol{u}' \equiv \eta_{\mathcal{H}}(\boldsymbol{u}) \tag{59}$$

The functions $\eta_{\mathcal{K}}(\boldsymbol{u})$ and $\eta_{\mathcal{H}}(\boldsymbol{u})$ are analogous to the vectors $\boldsymbol{X}\boldsymbol{w}^* = \boldsymbol{K}_X \in \mathbb{R}^M$ and $\boldsymbol{Y}\boldsymbol{v}^* = \boldsymbol{K}_Y \in \mathbb{R}^M$, and can be thought of as the 'decoded' functions. We will define the infinite-dimensional decoding similarity as the inner product:

$$S = \langle \eta_{\mathcal{K}}, \eta_{\mathcal{H}} \rangle_{L^2} = \int_{\mathcal{U}} \eta_{\mathcal{K}}(\boldsymbol{u}) \eta_{\mathcal{H}}(\boldsymbol{u}) p(\boldsymbol{u}) d\boldsymbol{u} \tag{60}$$

Then we have, using the definitions in eq. (58) and eq. (59),

$$S = \int_{\mathcal{U}} \int_{\mathcal{U}} \int_{\mathcal{U}} k(\boldsymbol{u}, \boldsymbol{u}')\eta(\boldsymbol{u}')\eta(\boldsymbol{u}'')h(\boldsymbol{u}, \boldsymbol{u}'')\, p(\boldsymbol{u})d\boldsymbol{u}\, p(\boldsymbol{u}')d\boldsymbol{u}'\, p(\boldsymbol{u}'')d\boldsymbol{u}''. \tag{61}$$

This integral measures the similarity between two neural systems f and g for a particular choice of decoding task function η. However, following the main text, we would like to take the expectation over some ensemble of decoding task functions η.

$$\mathbb{E}_\eta[S] = \mathbb{E}_\eta \langle \eta_{\mathcal{K}}, \eta_{\mathcal{H}} \rangle_{L^2} \tag{62}$$

Assuming the conditions to invoke Fubini's theorem regarding changing the ordering of integration are met, we have

$$\mathbb{E}_\eta[S] = \int_{\mathcal{U}} \int_{\mathcal{U}} \int_{\mathcal{U}} k(\boldsymbol{u}, \boldsymbol{u}')q(\boldsymbol{u}', \boldsymbol{u}'')h(\boldsymbol{u}, \boldsymbol{u}'')\, p(\boldsymbol{u})d\boldsymbol{u}\, p(\boldsymbol{u}')d\boldsymbol{u}'\, p(\boldsymbol{u}'')d\boldsymbol{u}''. \tag{63}$$

We now recognize this integral as this as the trace of a composition of integral operators,

$$\mathbb{E}_\eta[S] = \mathrm{Tr}[\mathcal{K}\mathcal{Q}\mathcal{H}] \tag{64}$$

where $\mathcal{Q} : L^2(\mathcal{U}) \to L^2(\mathcal{U})$ is the covariance operator associated with the Gaussian process covariance function q. This quantity is finite, since all of the operators involved are Hilbert-Schmidt and trace-class.

We now would like to show that the finite-dimensional notion of "decoding similarity" introduced in the main text (appropriately normalized) can be thought of as a plug-in estimator of the quantity eq. (64), and that this estimator converges to $\mathbb{E}_\eta[S]$ as $M \to \infty$.

A 'plug-in' estimator for this could be to use the Gram matrices for each kernel and matrix multiplication to approximate the integrals. First, we sample M examples from the input distribution $\boldsymbol{u} \sim P_{\boldsymbol{u}}$ and construct the Gram matrices $\boldsymbol{Q}_{ij} = q(\boldsymbol{u}_i, \boldsymbol{u}_j)$ and

$$\boldsymbol{K}_{ij} = k(\boldsymbol{u}_i, \boldsymbol{u}_j) = f(\boldsymbol{u}_i)^\top \boldsymbol{G}(f)^{-1} f(\boldsymbol{u}_j) \tag{65}$$

$$\boldsymbol{H}_{ij} = h(\boldsymbol{u}_i, \boldsymbol{u}_j) = g(\boldsymbol{u}_i)^\top \boldsymbol{G}(g)^{-1} g(\boldsymbol{u}_j). \tag{66}$$

Then we have, by the strong law of large numbers,

$$\int_{\mathcal{U}} \int_{\mathcal{U}} \int_{\mathcal{U}} k(\boldsymbol{u}, \boldsymbol{u}')q(\boldsymbol{u}', \boldsymbol{u}'')h(\boldsymbol{u}, \boldsymbol{u}'')\, p(\boldsymbol{u})d\boldsymbol{u}\, p(\boldsymbol{u}')d\boldsymbol{u}'\, p(\boldsymbol{u}'')d\boldsymbol{u}'' = \lim_{M\to\infty} \frac{1}{M^3} \sum_{ijk} K_{ij} Q_{jk} H_{ki}$$

or

$$\mathrm{Tr}[\mathcal{K}\mathcal{Q}\mathcal{H}] = \lim_{M\to\infty} \frac{1}{M^3} \mathrm{Tr}\, \boldsymbol{K}\boldsymbol{Q}\boldsymbol{H}. \tag{67}$$

However, the 'true' Gram matrices generated from the kernels in eq. (65) and eq. (66) in practice are also estimated, as the $N \times N$ regularization matrices $\boldsymbol{G}(f)$ and $\boldsymbol{G}(g)$ are potentially estimated from the same M samples from $P_{\boldsymbol{u}}$. We estimate $\boldsymbol{G}(f)$ and $\boldsymbol{G}(g)$ using the plug-in estimates of the $N \times N$ covariance matrices $\mathbb{E}[f(\boldsymbol{u})f(\boldsymbol{u})^\top]$ and $\mathbb{E}[g(\boldsymbol{u})g(\boldsymbol{u})^\top]$.

$$\boldsymbol{G}(\boldsymbol{X}) = \frac{\alpha}{M}\boldsymbol{X}^\top\boldsymbol{X} + \beta\boldsymbol{I} \xrightarrow[M\to\infty]{} \alpha\mathbb{E}[f(\boldsymbol{u})f(\boldsymbol{u})^\top] + \beta\boldsymbol{I} = \boldsymbol{G}(f) \tag{68}$$

$$\boldsymbol{G}(\boldsymbol{Y}) = \frac{\alpha}{M}\boldsymbol{Y}^\top\boldsymbol{Y} + \beta\boldsymbol{I} \xrightarrow[M\to\infty]{} \alpha\mathbb{E}[g(\boldsymbol{u})g(\boldsymbol{u})^\top] + \beta\boldsymbol{I} = \boldsymbol{G}(g) \tag{69}$$

More precisely, it can be shown that the empirical regularization matrices $\boldsymbol{G}(\boldsymbol{X})$ and $\boldsymbol{G}(\boldsymbol{Y})$ concentrates around $\boldsymbol{G}(f)$ and $\boldsymbol{G}(g)$ in operator norm: